

sors, microcontrollers, automotive power standard products, application-specific standard products, highly-integrated customised application-specific ICs (ASICs), cost-effective system-in-package (SiP) solutions and more. Plus, there are protocols such as PS15 and DSI to be complied with.

"The facility of e-call is being adopted by many automakers today. This technology automatically alerts emergency service providers to the location of an accident. With inter-vehicle communication, Ethernet-based parking-assistance solutions, crash-avoidance technologies, intelligent sensors for car battery monitoring, blind-spot detection, night vision and more, the industry is gearing up to the challenges and coming up with solutions that will have an overall impact on the safety of passengers," says Ganesh.

He adds, "2012 witnessed all cars having mandatory electronic stability control (ESC). ESC is the most significant safety feature since three-point seat belt. Another feature called intelligent speed adaptation is used to inform the driver about the speed limit for the road he is driving on and automatically reduce the vehicle's speed. The technology can very well benefit our country where every day one hears cases of high-speed collision with stationary vehicles on expressways."

Passenger comfort

Apart from Internet connectivity and the accompanying infotainment applications, automatic temperature control, personalisation of preferences, high-end displays, individualised media delivery, integrated display panel and electronic control for recliners are some of the features that contribute to

passenger comfort in high-end automobiles.

Lighting

With the growing popularity of light-emitting diodes (LEDs), electronic components are beginning to play a huge role in lights as well. LEDs add an element of intelligence to automotive lighting.

High-intensity discharge (HID) lamps provide up to three times more illumination than conventional halogen headlamps while reducing power consumption by roughly 30 per cent. These feature auto-levelling to reduce glare and improve visibility, and brightness or intensity adjustments according to ambient light. They are also more energy-efficient than previous technologies.

Body electronics

Body electronics addresses a range of applications inside vehicles, including door, window and seat control, entry control, HVAC, lighting and so on.

Several electronic components and sensors including a body control module, a central gateway, networking and communications technologies such as CAN, FlexRay, LIN and SAE J2602, 8-, 16- and 32-bit microcontrollers, motor control kits, and a broad range of analogue devices for power management and power switching are used. Special design, development and testing toolkits are used to design body electronics in an integrated and standardised fashion.

Instrument clusters

Simply put, instrument cluster is the dashboard that the driver sees. The goal of instrument cluster is to provide reliable and real-time information that

assists the driver. The in-vehicle network is used to draw information from the various ECUs embedded across the vehicle.

Today's information clusters are far from the simple mechanical meters of yore. These are completely digital panels built using low-power, automotive-grade embedded systems. They employ high-resolution displays, high-performance 2D/3D image rendering, turn-by-turn navigation, dynamic human-machine interface that can be adapted to day and night conditions, information gateways and in-built security features.

Most instrument clusters are reconfigurable, allowing the driver to choose from different preset configurations. For example, the driver can choose to view the data in traditional gauge format or in a trendy new fashion. Some vehicles even allow new formats to be downloaded as applications from the Internet.

Summing up

The number, capabilities and configurability of electronic components might vary from low-end to high-end cars, but surely almost all of today's vehicles come with a mandatory dose of electronics. This is true in India too. In fact, there are many leading companies offering products and services in this space, including Bosch Diagnostics, Freescale Semiconductor, NXP, Texas Instruments, Renesas, Visteon, TCS, HCL, Wipro, KPIT Cummins and Persistent Systems. These firms extend their offerings not just to the Indian automotive industry but the global market as well! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

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